William Schober

PhD student in Quantum Computing

Nationalities	USA and Austria
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Education

2022 - 2027 **PhD in Informatics**, Università della Svizzera italiana, Lugano, Switzerland
Thesis: *Equational theories for quantum circuits, with application to circuit optimization.*
Supervisor: Prof. Stefan Wolf

2019 - 2021 **MSc in Artificial Intelligence**, Università della Svizzera italiana, Lugano, Switzerland
Thesis: *Geometry of quantum state space: an investigation by convexity and unitary orbits.*
Supervisor: Prof. Stefan Wolf

2014 - 2017 **BSc in Physics and Mathematics**, University of Massachusetts Amherst, Amherst, MA, USA
Undergraduate Research: Analyzed collision data from the Large Hadron Collider using binary classification neural networks written in ROOT, a C++ derived programming language, on a Linux platform. Supervisor: Prof. Stéphane Willocq

Publications

2026 **W. Schober** and S. Wesley. A New Method For Manipulating Circuits, Application To Quantum Adders. arXiv

2026 **W. Schober** and S. Wesley. A Complete Equational Theory for Quantum Circuits with Generalized Control. *Electronic Proceedings in Theoretical Computer Science* 456, pp. 283-349.
Best Student Paper Award at the 23rd International Conference on Quantum Physics and Logic (QPL 2026). arXiv DOI

2024 **W. Schober**. Extended Quantum Circuit Diagrams. arXiv

Teaching and Supervision

2022 - 2026

Teaching assistant, Computer Science courses, Università della Svizzera italiana

Led exercise sessions and designed and graded homework and exams, working in teams of 1-4 TAs for classes of 20-80 students.

Information & Physics	MSc	Spring 2026
Software Atelier 1: Fundamentals of Informatics	BSc	Fall 2025
Information & Physics	MSc	Spring 2025
Software Atelier 1: Fundamentals of Informatics	BSc	Fall 2024
Information & Physics	MSc	Spring 2024
Programming Fundamentals 1	BSc	Fall 2023
Information & Physics	MSc	Spring 2023
Numerical Algorithms	MSc	Fall 2022

2025 - 2026

Thesis supervisor, Computer Science students, Università della Svizzera italiana

- **Samuele Galbusera** BSc thesis, 2026
  Does provable absence of barren plateaus imply classical simulability?
- **Mark Sobolev** MSc thesis, 2025
  On Qubit Routing in Quantum Computer Architectures.

Talks

Oct 2026

A Complete Equational Theory for Quantum Circuits with Generalized Control. Seminar talk, École de technologie supérieure quantum seminar series, Quebec. Video

Aug 2026

A Complete Equational Theory for Quantum Circuits with Generalized Control. Conference talk, QPL 2026, Amsterdam. Video

Jan 2026

Towards a Hierarchical Quantum Circuit Language. Workshop talk, PLanQC 2026, Rennes. Video

Posters

Feb 2025

W. Schober. *Extended quantum circuit diagrams.* QIP 2025, Raleigh.

Sep 2024

W. Schober. *Circuits as a self-contained graphical language.*TQC 2024, Okinawa.

Jul 2024

W. Schober. *Circuits as a self-contained graphical language.* QPL 2024, Buenos Aires.

Jan 2024

W. Schober. *Extended Quantum Circuit Diagrams.* QIP 2024, Taipei.

Last updated November 2026.